

Machine Learning Assignment Report - Sri Lanka Foreign Tourist Arrivals Prediction

Source Code Git Hub Link -

https://github.com/DilshanPathegamage123/SriLanka_Tourism_Prediction_ML_Project

1. Problem Definition & Dataset Collection

Problem Definition

Problem: Sri Lanka's tourism sector experiences significant fluctuations due to seasonality, country-specific demand patterns, economic shifts, and unexpected global events such as the COVID-19 pandemic. These fluctuations create high volatility and structural breaks in arrival patterns, making it difficult for policymakers, tourism authorities, hotels, and airlines to accurately anticipate future foreign tourist arrivals.

Traditional statistical forecasting approaches often struggle to model:

- Strong seasonal cycles
- Non-linear growth and recovery patterns
- Country-level heterogeneity
- Structural disruptions such as pandemics

As a result, there is a need for a more adaptive and data-driven forecasting approach.

Develop a machine learning model that captures temporal dependencies, seasonality, country-specific behavior, structural shocks (COVID), and market evolution to accurately forecast monthly foreign tourist arrivals from each source country to Sri Lanka.

Data Collection

Data Source:

- Sri Lankan Tourism Data from - <https://ldflk.github.io/datasets/browse/interactive-browser/>
- Sri Lanka Tourism Development Authority (SLTDA) publicly available reports - <https://www.slttda.gov.lk/en/statistics>

Collection Method:

Downloaded the datasets below for each year (from 2019 to 2024), compiled them into one dataset, and applied preprocessing techniques.

- Tourist arrivals by country and month
- Top 10 source markets with market share
- Accommodation capacity by province
- Attraction revenue and visitor counts for each travel location in Sri Lanka.
- Total Tourist arrivals by month

Dataset Size:

- **Raw data:** JSON files per year × 6 years = 30 files
- **Final processed dataset:** 13,632 rows × 47 features
- **Time period:** January 2019 - December 2024 (6 years)

Features and Target Variable:

Target: Tourist_Arrivals - monthly tourist arrivals from each country

Features (46 total):

- **Temporal:** Year, Month_Num, Month_Sin, Month_Cos, Quarter indicators
- **Country-specific:** Country_Code, Is_Top10, Market_Share, Is_Consistent_Top10
- **Lag features:** Lag_1_Month (Arrivals last month), Lag_2_Month (Arrivals last 2 month), Lag_3_Month, Lag_6_Month, Lag_12_Month
- **Rolling statistics:** Rolling_Mean_3/6/12, Rolling_Std_3/6/12
- **Growth rates:** Growth_3M, Growth_12M, Prev_Year_Same_Month_Ratio
- **Accommodation:** Total_Rooms, Avg_Rooms_per_Province, Std_Rooms_per_Province
- **Attractions:** Total_Foreign_Visitors_Attractions, Total_Foreign_Income
- **COVID indicators:** Is_COVID_Year, Is_Post_COVID, COVID_Severity
- **Interaction features:** COVID_Season_Interaction, Top10_COVID_Effect

Preprocessing Steps

Data Integration: Merged all JSON files into consolidated CSV files

Cleaning:

- Handled missing values using forward- and backward-fill.
- Interpolated accommodation data gaps.
- Standardized place names in attraction data.

Encoding:

- Label encoded country names and the month names.
- One-hot encoded quarters

Normalization: Applied StandardScaler to numeric features

Feature Engineering: Created 46 features, including lags, rolling statistics, and interaction terms

Ethical Considerations: Used publicly available aggregated data with no personal or sensitive information. No individual tourist data was collected.

2. Selection of Machine Learning Algorithm

Algorithm Selected: Random Forest Regressor

Why choose Random Forest

- **Handles non-linearity:** Tourist arrivals have complex seasonal patterns, COVID crashes, and recovery trends that linear models cannot capture.
- **Robust to outliers:** Tourism data contains extreme values (COVID zeros, seasonal peaks up to 50,000+) - Random Forest handles these naturally.
- **Built-in interpretability:** Provides feature importance scores for XAI requirements (SHAP analysis).
- **No distribution assumptions:** Works well with skewed tourism data without requiring normalization.
- **Handles mixed data types:** Can process numerical (arrivals), categorical (country codes), and binary (COVID indicators) features together.

How it differs from standard models

Model	Limitation	Why Random Forest is Better
Linear Regression	Assumes linear relationships	Tourism patterns are highly non-linear (seasonal peaks, COVID crash)

Decision Tree	High variance, overfitting	Ensemble of trees reduces overfitting
k-NN	Sensitive to feature scaling, curse of dimensionality	Handles high-dimensional data well (46 features)
Logistic Regression	Classification only	The regression task requires continuous output
SVM	Poor interpretability, sensitive to parameters	Built-in feature importance for explainability

3. Model Training and Evaluation

Data Split Strategy

Time-based split (preserving temporal order)

- Training: 2019-2022 (9,036 samples) - Pre-COVID, COVID, recovery period
- Validation: 2023 (2,280 samples) - Post-COVID recovery validation
- Test: 2024 (2,316 samples) - Holdout for final evaluation

A standard split would leak future information into training. Time-series requires chronological ordering.

Hyperparameter Tuning

Search Space:

```
param_dist = {
    'n_estimators': randint(100, 500),    # Number of trees
    'max_depth': randint(5, 20),         # Tree depth
    'min_samples_split': randint(2, 20),  # Samples to split node
    'min_samples_leaf': randint(1, 10),   # Min samples in leaf
    'max_features': ['sqrt', 'log2', None] # Features per split
}
```

Best Parameters Found:

Parameter	Value	Interpretation
n_estimators	459	High ensemble size for stability
max_depth	7	Moderate depth prevents overfitting

min_samples_leaf	8	Prevents learning noise
min_samples_split	5	Balanced splitting
max_features	None	All features are considered at splits

Performance Metrics

Validation Set (2023):

Metric	Value	Interpretation
MAE	137	Average error of 137 tourists per country-month
RMSE	723	Higher RMSE indicates some large errors
R ²	0.928	92.8% variance explained

Test Set (2024):

Metric	Value	Interpretation
MAE	178	Slightly higher error (new patterns in 2024)
RMSE	991	Some large prediction errors
R ²	0.920	92% variance explained - excellent fit

Results Analysis

R² = 0.92 (Test Set)

- Excellent fit - Model captures 92% of patterns in tourist arrivals.
- Only 8% variance unexplained (noise, external factors).
- Slight drop from 0.928 → 0.920 indicates good generalization.

MAE = 178 (Test Set)

- On average, predictions are off by 178 tourists per country-month.
- Context: Monthly arrivals range from 0 to 50,000+
- For major markets (India: 30,000+), this is <1% error.

- For tiny markets (Bhutan: 50), this is 356% error - a clear limitation.

RMSE = 991 (Test Set)

- RMSE > MAE (5.6× larger) confirms the presence of large errors.
- Some predictions are off by thousands, not hundreds.
- These large errors come from major markets (India, UK, Russia), where absolute numbers are high.

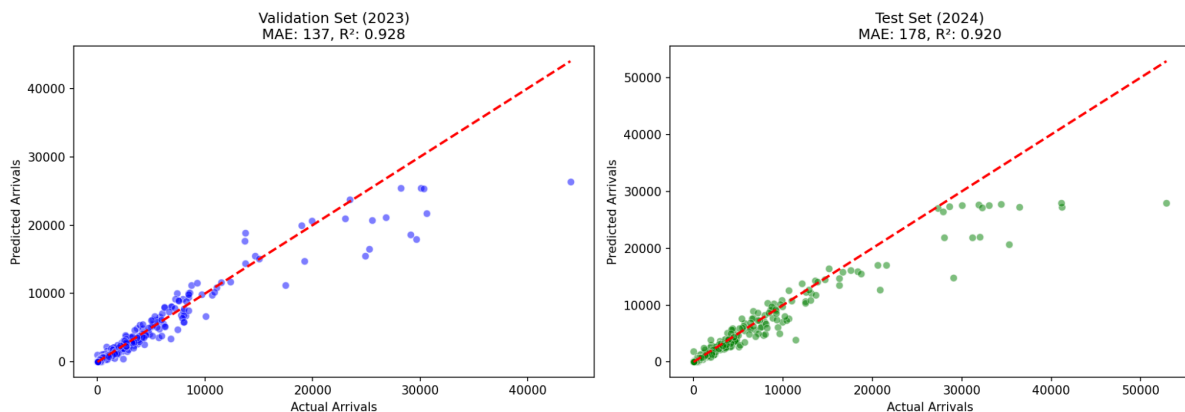
Why performance dropped in 2024 (Test Set):

- Arrivals from China doubled.
- Russia became the 2nd largest market (geopolitical shift).
- The Netherlands entered the Top 10 (new emerging market).
- Training data (2019-2022) couldn't capture these new patterns.
- Model extrapolated based on old relationships, which changed.

Visualization Results

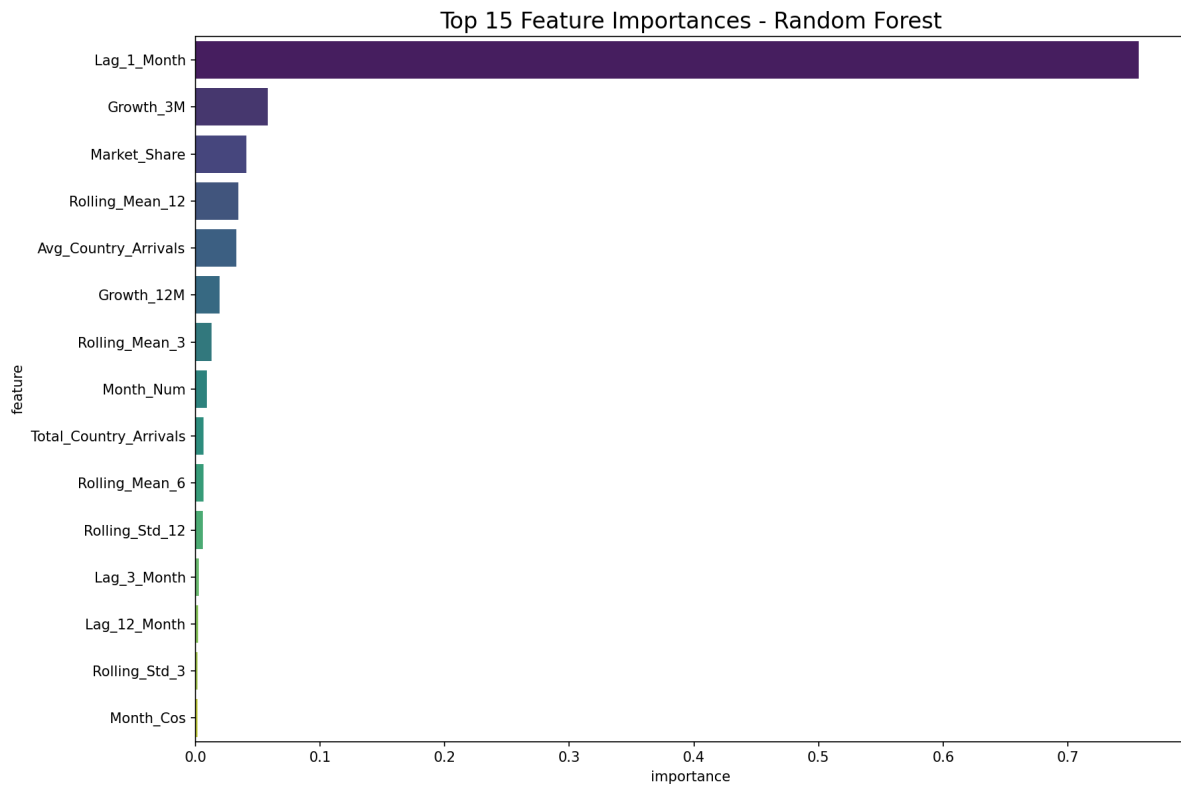
Actual vs Predicted Plot:

- Validation set shows tight clustering around the diagonal
- The test set shows slightly more scatter but maintains a strong correlation
- Red line (perfect prediction) closely followed by actual points



Feature Importance Plot:

- Lag_1_Month dominates (76% importance)
- Growth_3M (5.8%) and Market_Share (4.1%) next most important
- Confirms tourism is highly autocorrelated



Feature importance analysis...

Collapse Output

Top 15 most important features:

feature	importance
Lag_1_Month	0.757031
Growth_3M	0.058179
Market_Share	0.041066
Rolling_Mean_12	0.034530
Avg_Country_Arrivals	0.033025
Growth_12M	0.019225
Rolling_Mean_3	0.012800
Month_Num	0.009386
Total_Country_Arrivals	0.006514
Rolling_Mean_6	0.006227
Rolling_Std_12	0.006040
Lag_3_Month	0.002897
Lag_12_Month	0.002072
Rolling_Std_3	0.001710
Month_Cos	0.001490

4. Explainability & Interpretation

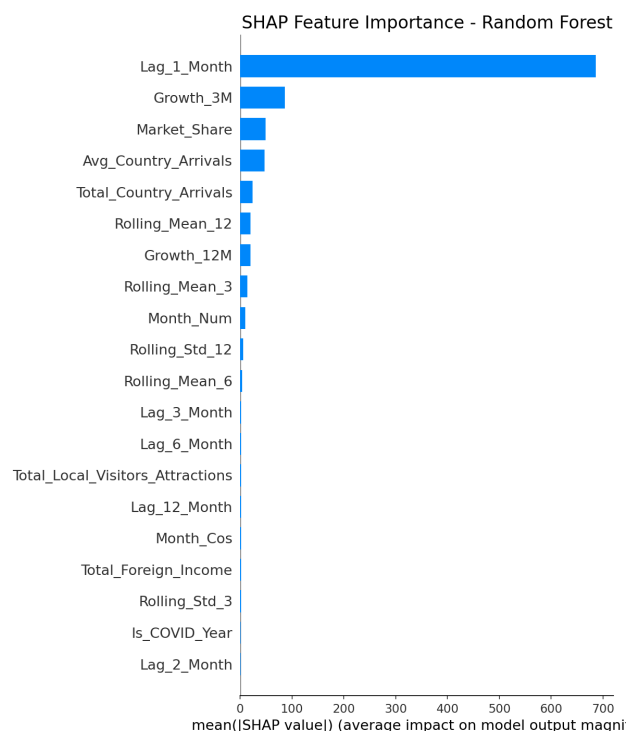
Method Applied: SHAP

What the Model Learned:

- Temporal dependencies: Strong autocorrelation (Lag_1_Month dominates)
- Country characteristics: Market Share and historical averages differentiate countries
- Seasonality: Month_Num and Month_Cos capture periodic patterns
- COVID impact: COVID_Severity appears in the top 20 features

Most influential features:

1. Lag_1_Month Dependence:
 - Positive relationship: Higher previous month → Higher prediction
 - Color shows Rolling_Mean_3 interaction
 - Validates tourism momentum
2. Growth_3M Dependence:
 - Positive growth increases predictions
 - Negative growth (decline) reduces predictions
 - Interaction with rolling averages shows context matters
3. Market_Share Dependence:
 - A larger market share makes higher predictions
 - Non-linear effect at extremes
 - Countries with >4% share show amplified impact



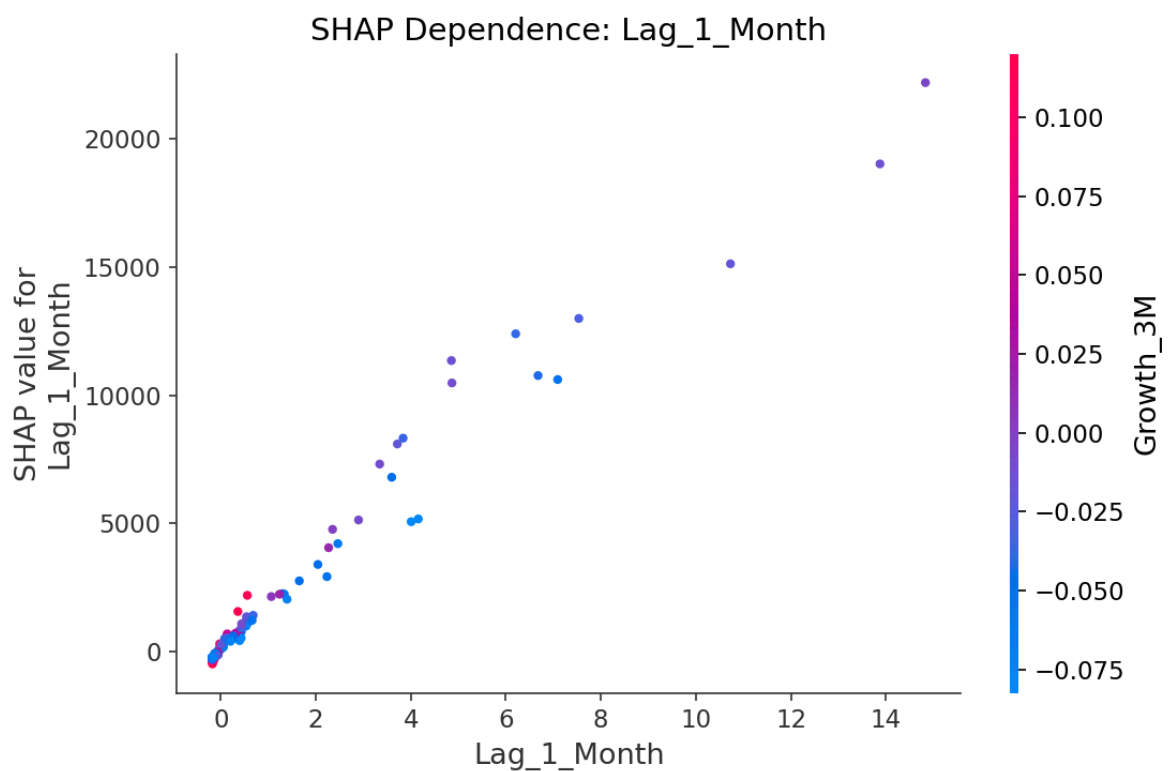
Alignment with Domain Knowledge

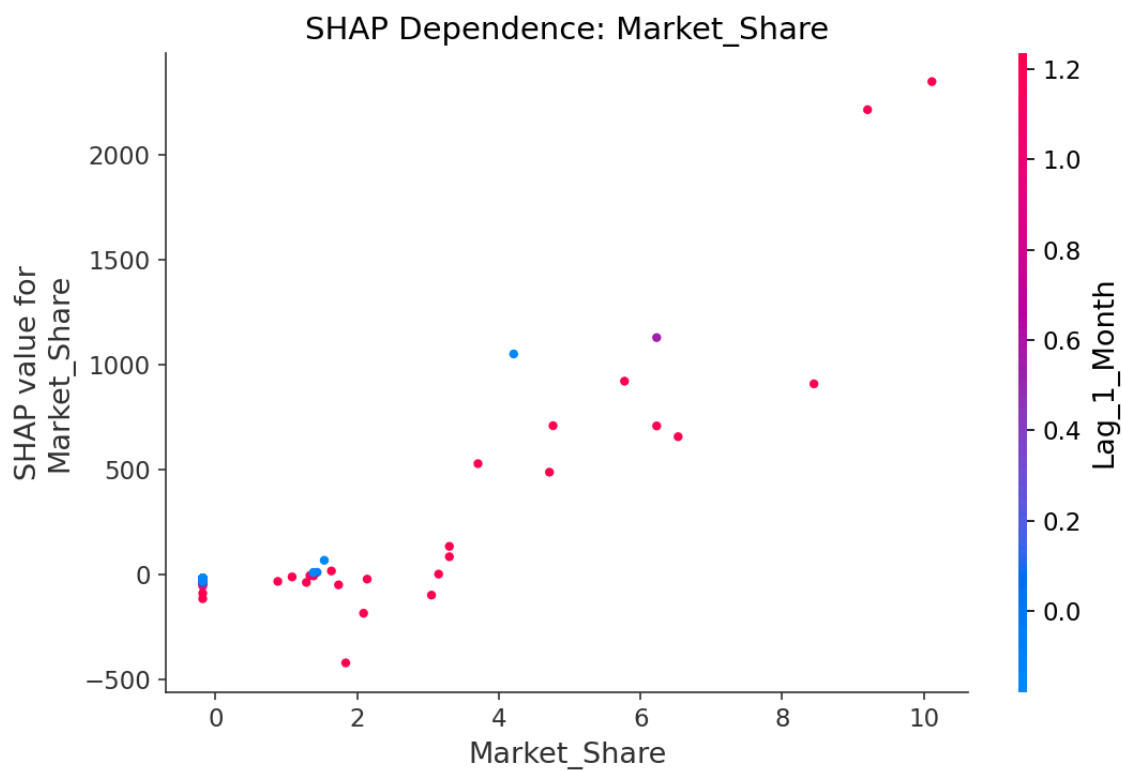
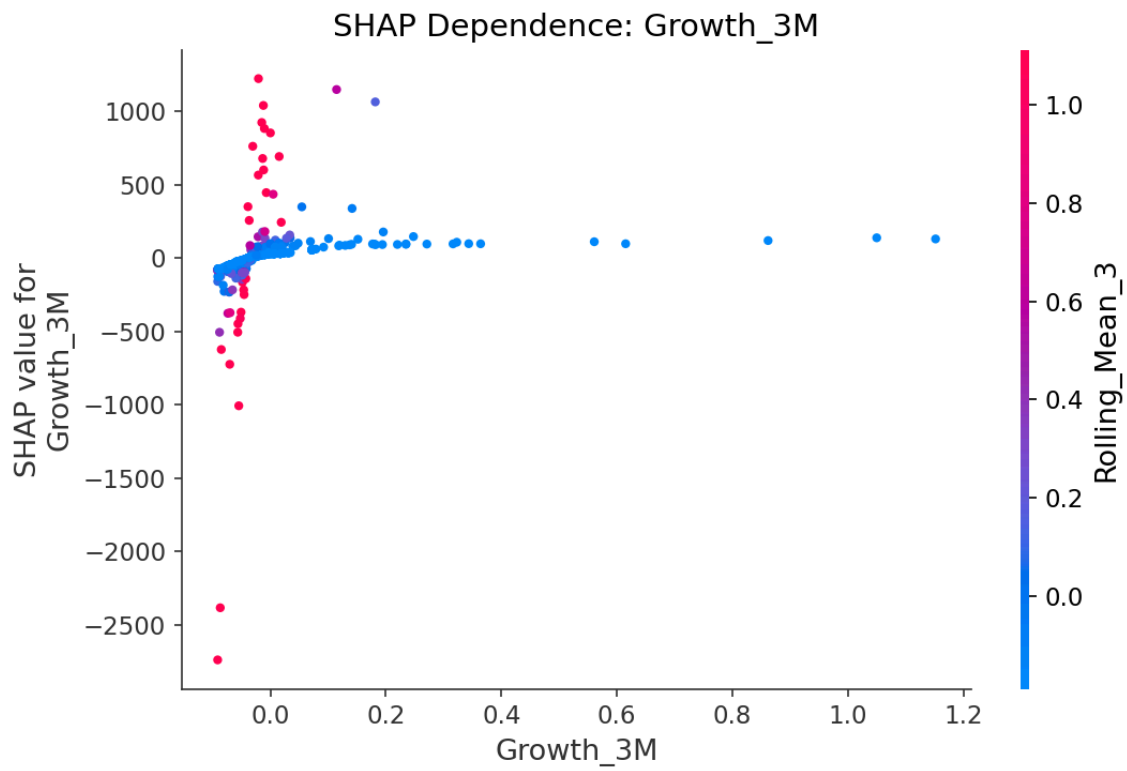
Confirmed patterns:

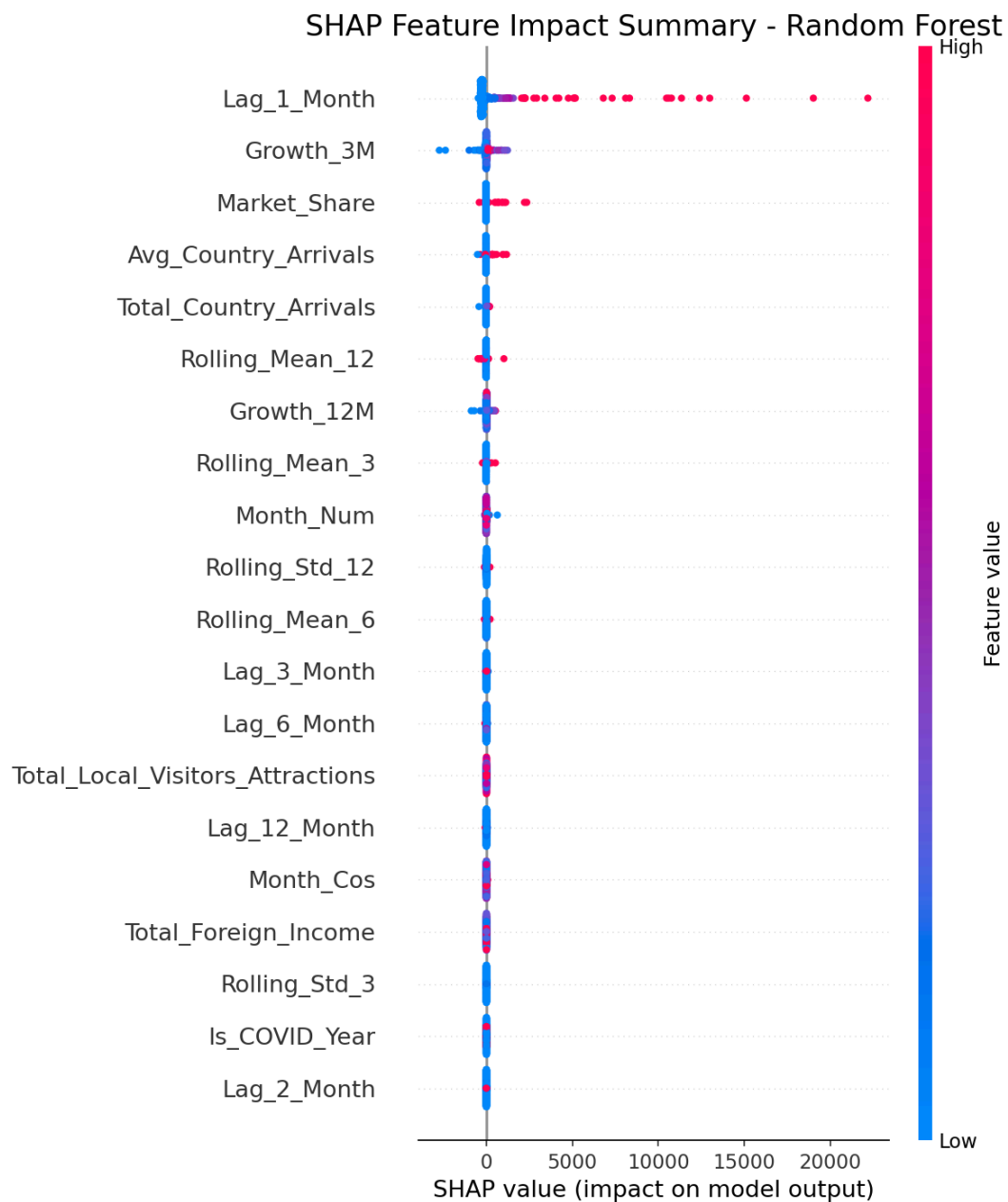
- Tourism is highly seasonal (matches industry knowledge)
- COVID caused a structural break (captured by severity indicators)
- Major markets (India, UK, Russia) dominate (Market_Share important)
- Recovery patterns show momentum (lag features are important)

Unexpected findings:

- Accommodation capacity is less important than expected
- Attraction of visitors with minimal impact.



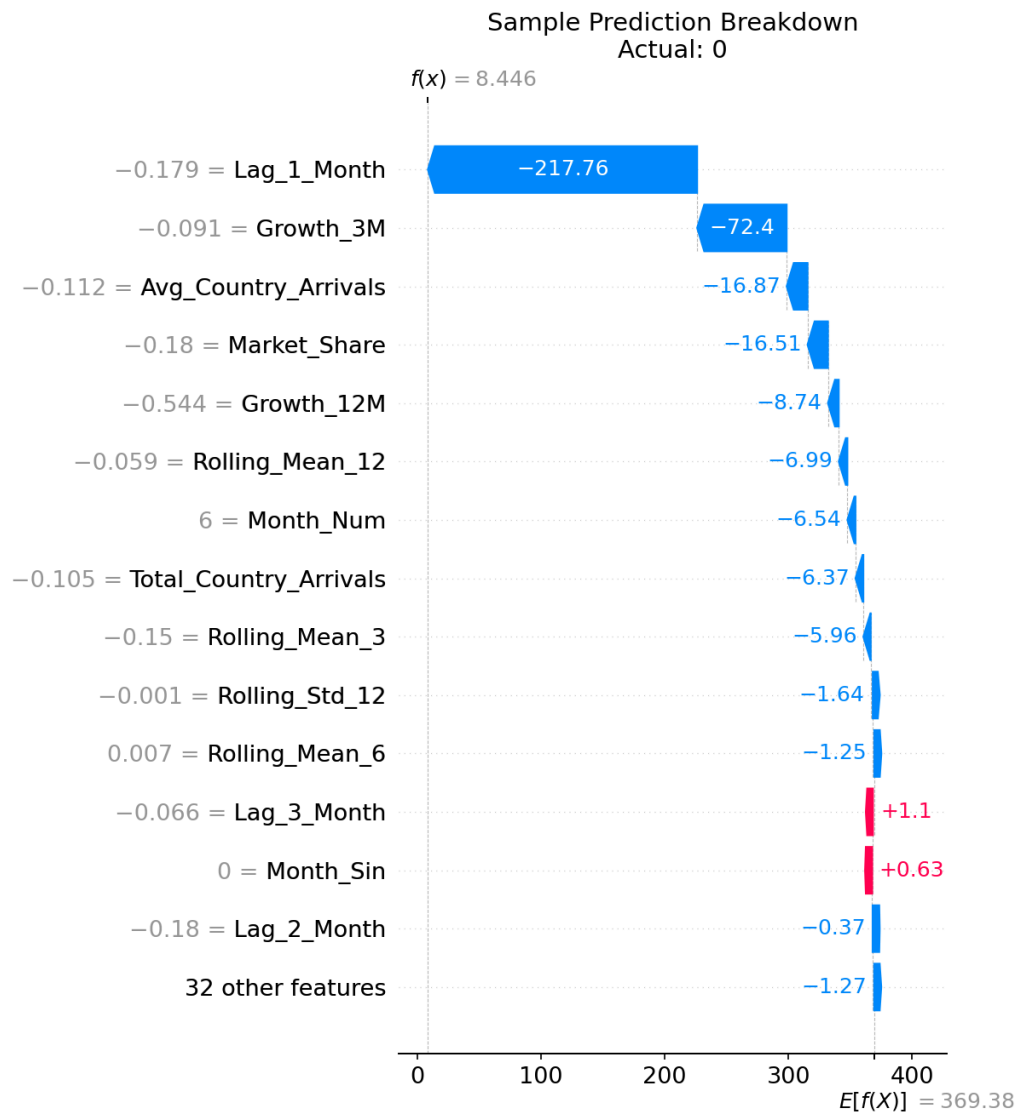




Sample Waterfall Plot Analysis

For a typical prediction:

- Base value: ~370 arrivals
- Lag_1_Month: +369 contribution (dominant)
- Growth_3M: +26 contribution
- Avg_Country_Arrivals: +4 contribution
- Market_Share: -14 contribution (this case had a lower share)



5. Critical Discussion

Model Limitations Identified

1. Tiny Market Prediction Failure

- For countries with <100 monthly arrivals, the error often exceeds the actual value
- Example: Bhutan (50 actual) → predicted 187 (274% error)
- Cause: Sparse data, high volatility in small numbers

2. Future Prediction Uncertainty

- For 2027+ predictions, lag features must be estimated
- Each estimated feature adds uncertainty
- Confidence intervals widen significantly for future years

3. External Factors Not Captured

- Exchange rates, airline capacities and routes, competitor pricing
- Geopolitical events (Russia-Ukraine war impact)
- Economic conditions in source countries

4. Temporal scope: Only 6 years of data (2019-2024 includes the abnormal COVID period)

Data Quality Issues

Issue	Issue	Mitigation
Inconsistent 2019 province names	Data loss	Manual mapping and aggregation
Missing attraction data 2020-2021	COVID years incomplete	Interpolation used
Zero values during COVID lockdowns	Extreme values affecting statistics	COVID severity indicators added
Top10 markets only partial coverage	Other countries lack market share data	Is_Top10 flag created

Risks of Bias or Unfairness

- Market concentration bias: Model better for top markets (more data available)
- Recency bias: 2024 patterns weighted heavily (test set shows drift)
- COVID distortion: 2020-2021 patterns may not represent normal tourism
- Small country bias: Predictions for minor markets are less reliable (fewer historical patterns)

Real-World Impact & Ethical Considerations

Positive impacts:

- Tourism planning optimization
- Resource allocation efficiency
- Marketing budget optimization

Risks:

- Over-reliance on predictions could lead to poor decisions
- The model doesn't account for unexpected events (crises, pandemics)
- May perpetuate existing market biases (focus on top markets)

Ethical use guidelines:

- Always combine with human judgment
- Update the model regularly with new data
- Monitor prediction errors continuously
- Don't use for critical policy decisions without expert review